

Quiz 3

ECON 320 – Introduction to Mathematical Economics

Instructor: Muhebullah Karimzada

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Part A – Multiple Choice Questions (1 point each)

1. If $f(x)$ has a local maximum at $x = a$, then
 - (a) $f'(a) > 0$
 - (b) $f'(a) = 0$ and $f''(a) > 0$
 - (c) $f'(a) = 0$ and $f''(a) < 0$
 - (d) $f'(a) < 0$
2. The second-order condition for a *minimum* of a differentiable function $f(x)$ is
 - (a) $f'(x) = 0$ and $f''(x) < 0$
 - (b) $f'(x) = 0$ and $f''(x) > 0$
 - (c) $f''(x) = 0$
 - (d) $f'(x) > 0$
3. If $f(x) = x^3 - 6x^2 + 9x$, the function attains a minimum at
 - (a) $x = 1$
 - (b) $x = 2$
 - (c) $x = 3$
 - (d) $x = 0$

4. The profit function $\pi(Q) = 120Q - 5Q^2$ reaches its maximum when marginal revenue equals
- (a) 120
 - (b) zero
 - (c) 5
 - (d) average cost
5. The function $y = x e^{-x}$ has its maximum value when
- (a) $x = 0$
 - (b) $x = 1$
 - (c) $x = 2$
 - (d) $x = 1/e$
6. If $f(x) = a - bx^2$ with $a, b > 0$, then $f(x)$ is
- (a) strictly concave
 - (b) strictly convex
 - (c) linear
 - (d) neither concave nor convex
7. The necessary condition for an interior extremum of $f(x)$ is that
- (a) $f'(x) \neq 0$
 - (b) $f''(x) = 0$
 - (c) $f'(x) = 0$
 - (d) $f''(x) > 0$

Part B – Analytical Problems (13 marks total)

Question 1 (4 points)

A firm's profit is given by

$$\pi(Q) = 100Q - 4Q^2 - 20.$$

1. Derive the first-order condition.
2. Find the profit-maximizing output Q^* .
3. Compute the maximum profit.
4. Verify the second-order condition.

Question 2 (4 points)

Let the utility of consumption be

$$U(x) = 10x - 0.5x^2.$$

1. Find x^* that maximizes utility.
2. Check whether this is a maximum.
3. Evaluate $U(x^*)$.
4. Explain the economic meaning of the result.

Question 3 (5 points)

The total revenue of a monopolist is

$$R(P) = 500Pe^{-0.1P}.$$

1. Derive the price P^* that maximizes total revenue.
2. Compute the corresponding maximum revenue $R(P^*)$.
3. Verify that the second-order condition holds.
4. Discuss why $R(P)$ first rises and then falls as P increases.